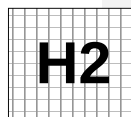


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST ANDREW'S JUNIOR COLLEGE
2016 JC2 Preliminary Examinations

H2 BIOLOGY

9648/3

Paper 3: Applications and Planning Question

Tuesday

13th September 2016

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your civics group, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,

1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/12
2	/17
3	/11
Total	/40
4 (Planning)	/12

This document consists of **13** printed pages.

[Turn over

Answer all questions.

QUESTION 1

Tissue plasminogen activator (TPA) is a blood protein that scavenges and dissolves blood clots that may form in the final stages of the healing process. TPA is primarily used in heart patients or others suffering from poor circulation to prevent the development of clots that can be life-threatening. Hence, microbially produced TPA is in high demand.

Fig. 1.1 shows a commercially available plasmid that can be used to clone a synthetic TPA gene into *E. coli*. It contains two selectable markers which confer resistance to the antibiotics vancomycin (Van^R gene) and ampicillin (Amp^R gene) respectively.

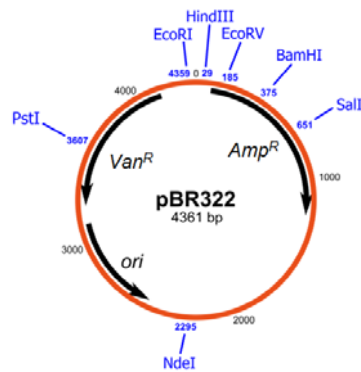


Fig. 1.1

(a)(i) Describe the natural function of restriction enzymes.

.....[1]

(ii) *BamHI* was used to produce recombinant plasmids containing TPA gene. Describe and explain the selection procedure that would allow for selection of bacteria which contain the recombinant plasmids.

.....[5]

(iii) The TPA gene which is inserted into the plasmid can be artificially synthesised. This synthetic TPA gene inserted into the plasmid may possess a different nucleotide sequence to the actual human gene. Explain one reason why this is so.

.....

[2]

(b) Discuss one problem of producing eukaryotic proteins using a prokaryotic host and how it can be overcome.

.....

[3]

(c) Two DNA samples of insulin gene were obtained using different methods:

1st method: reverse transcription of insulin mRNA to produce cDNA

2nd method: extraction of genomic DNA followed by selective amplification of the insulin gene using PCR

The two DNA samples were loaded into **different** wells and separated via **gel electrophoresis**. Draw the band(s) you would expect to see for the DNA sample obtained using the 2nd method in Fig 1.2.

[1]

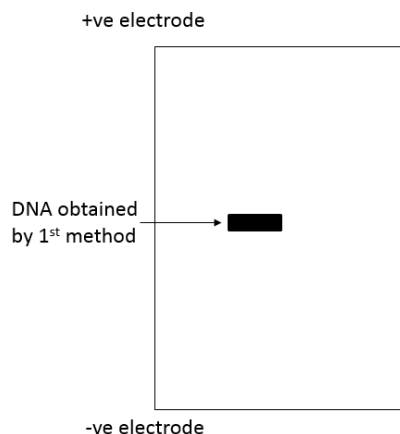


Fig. 1.2

[Q1 Total: 12]

QUESTION 2

Individuals born with severe combined immune deficiency (SCID) have no defence against common infections. This disease is caused by the possession of recessive alleles.

Fig.2.1 shows the pedigree which traces the inheritance of SCID in a family.

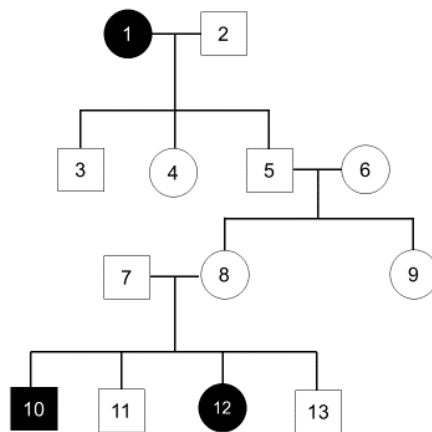


Fig. 2.1

(a)(i) Explain if the SCID tracked in Fig. 2.1 is autosomal or X-linked.

.....[1]

(ii) Explain the genetic basis behind the susceptibility to common infections in this mode of SCID.

.....[3]

(iii) Explain why SCID is a suitable candidate for treatment using gene therapy.

.....[2]

Commented [WU1]: susceptibility

Gene therapy is carried out for individual 1 (shown in Fig.2.2). The essential steps are summarized in Fig.2.2.

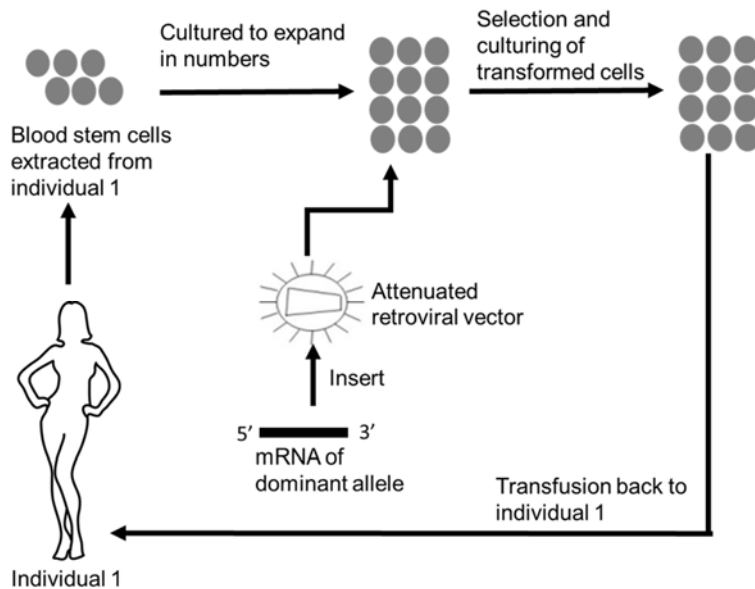


Fig.2.2

(iv) **Explain-Give two reasons** why blood stem cells are chosen to receive the dominant allele.

.....

[2]

(v) Suggest why the *ex vivo* method was used in this case.

.....
[1]

(vi) The gene therapy treatment described in Fig.2.2 turned out to be ineffective in treating individual. Suggest two reasons why.

.....

[2]

Commented [NYHE2]: insert "two reasons"? If we are looking for both points. I scared students will just write one of them and elaborate thinking it will get them 2 marks already.

(vii) Explain why individuals treated **successfully** with gene therapy do not pass on the dominant allele to their children.

.....
[1]

(viii) Discuss two ethical considerations for the use of gene therapy.

.....

[2]

It is possible to diagnose for SCID in the individuals shown in Fig. 2.1, by detecting known RFLP regions found **within** the gene.

(ix) Explain how the Human Genome Project facilitates this process.

.....
[1]

In SCID, a mutation alters one base of the coding sequence, resulting in a loss of a restriction site. The gene responsible for SCID is isolated from **individual 8** of Fig. 2.1. Restriction digestion with *MspI* enzyme results in 0.47 kb, 1.834 kb and 2.304 kb fragments.

(x) The lines in Fig. 2.3 represent the RFLP locus on the homologous chromosomes. Draw the restriction map showing the **cut sites** of *MspI* enzyme, **using arrows**, for individual 8. The **size** of the fragments between cut sites should be indicated.

.....[1]



Fig. 2.3

A DNA probe is added which resulted in the detection of only the 1.834 kb and 2.304 kb bands.

(xi) Draw the suggested position of the probe in Fig. 2.3.

.....[1]

[Q2 Total: 17]

QUESTION 3

In Singapore, orchids of superior quality are selected for mass production through tissue culture.

Tissue culture is a collection of techniques used to grow plant cells, tissues, or organs under sterile conditions on a nutrient culture medium.

- (a) Equal masses of orchid plant callus were cultured on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 3.1.

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on the callus
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

Table 3.1

With reference to Table 3.1,

- (i) describe the effects of auxin and cytokinin on the callus.

.....

.....

.....

.....

.....

.....[3]

- (ii) suggest what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets.

.....

.....[1]

(b) Explain why plantlets grown from a callus in tissue culture are considered to be clones of each other.

.....

[2]

(c) Tissue culture is also used to grow genetically engineered plant cells into whole plants. An example are crops which are genetically engineered to carry the Bt toxin gene from the bacteria *Bacillus thuringiensis*.

Explain the benefit of incorporating a *Bt* gene into a crop plant.

.....

[2]

(d) The genetically modified (GM) salmon was created in 1992 by Emeritus Professor Hew Choy Leong from the department of biological sciences at the National University of Singapore.

Explain why this development of GM salmon is supported by some.

.....

[3]

[Q3 Total: 11]

4. Planning question

You are required to plan, but not carry out, an investigation into the effect of different temperatures on the rate of respiration in crickets.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- Crickets
- Manometer with scale (internal radius of manometer is 1mm)
- Manometer fluid
- 2 rubber bungs with delivery tubes
- 2 boiling tubes
- Zinc gauze platform (to provide platform to support the crickets)
- Glass beads
- Clamps and stands
- Stop watch
- Soda lime solution
- Thermometer
- Supply of water at 50°C
- Tap water / ice
- Weighing balance
- A variety of different sized beakers, measuring cylinders, and syringes for measuring volumes

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure,
- a description of the method used, including the scientific reasoning behind the method and any recommended safety measures,
- an explanation of dependent and independent variables involved,
- relevant and clearly labeled diagrams,
- how you will record your results and ensure they are accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms.

[Q4 Total: 12]

[illegible]

SAJC / H2 Biology 9648/3 JC2 Prelim 2016

This image shows a full page of a document template designed for handwriting practice or general note-taking. It consists of approximately 28 evenly spaced horizontal dotted lines across the entire width of the page. There are no margins, headers, footers, or other markings present.

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** Discuss the advantages and limitations of the Polymerase Chain Reaction. [8]
 (b) Explain how gel electrophoresis is used to analyse DNA. [5]
 (c) Discuss the negative implications of Genetically Modified Organisms. [7]

[Total: 20]